

Challenge (Habanero)

Q1. Simplify $\sqrt{8} + \sqrt{2}$

- (A) $2\sqrt{2} + \sqrt{2}$
- (B) $3\sqrt{2}$
- (C) 4
- (D) $2\sqrt{3}$
- (E) $\sqrt{10}$

Q2. Simplify $\sqrt{12} - \sqrt{3}$

- (A) $\sqrt{9}$
- (B) $2\sqrt{3}$
- (C) $\sqrt{15}$
- (D) 4
- (E) $\sqrt{12} - \sqrt{3}$

Q3. A chef needs to multiply $\sqrt{6}$ and $\sqrt{2}$ to calculate a recipe. What is the result?

- (A) $\sqrt{8}$
- (B) $\sqrt{12}$
- (C) $2\sqrt{3}$
- (D) $\sqrt{5}$
- (E) $\sqrt{3}$

Q4. Simplify $\frac{\sqrt{20}}{\sqrt{5}}$

- (A) 2
- (B) $\sqrt{4}$
- (C) $\sqrt{20}$
- (D) 4
- (E) $\sqrt{2}$

Q5. Simplify $(\sqrt{3} + \sqrt{2})^2$

- (A) $5 + 2\sqrt{6}$
- (B) $2 + 2\sqrt{6}$
- (C) $3 + 2\sqrt{6}$
- (D) $3 + \sqrt{6}$
- (E) $\sqrt{5}$

Q6. A construction worker needs to find the length of a side of a triangular roof. The side can be represented by $\sqrt{8} + \sqrt{2}$. Simplify to find the length.

- (A) $2\sqrt{2} + \sqrt{2}$
- (B) $3\sqrt{2}$
- (C) 4
- (D) $2\sqrt{3}$
- (E) $\sqrt{10}$

Q7. Simplify $\sqrt{18} - \sqrt{8}$

- (A) $2\sqrt{2} - \sqrt{2}$
- (B) $\sqrt{9}$
- (C) $\sqrt{15}$
- (D) 4
- (E) $\sqrt{10}$

Q8. Simplify $(\sqrt{5} - \sqrt{3})(\sqrt{5} + \sqrt{3})$

- (A) 2
- (B) $5 - 3$
- (C) $5 + 3$
- (D) $\sqrt{2}$
- (E) $2\sqrt{15}$

Open Ended Questions (FRQ)

Q9. Simplify the expression $(\sqrt{6} + \sqrt{3})(\sqrt{2} - \sqrt{3})$ and provide your work.

Q10. A recipe calls for $\sqrt{24}$ cups of flour. Simplify the radical and then multiply by 2 to find the amount needed for a double recipe. Show all steps.

Chef's Tips

- Provide clear instructions and examples before the test.
- Allow students to use calculators for checking purposes only.
- Remind students to simplify all radicals and show work for free response questions.